

Slot B

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO1: Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)

Assessment Tool Weightage: 35%

QUESTIONS: 5

Total Marks: 25

Annexure A

SHORT ANSWER TEST

Code of Conduct:

Iv. Hussain 192524179 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Iv. Hussain

Annexure A

Short Answer Test

Question 1 (5 Marks)

Suppose a modern Operating System supports multiple users and processes with shared resources like CPU, memory, and I/O devices.

- Classify different types of Operating Systems suitable for this scenario and justify your choice.
- Draw the structure of an Operating System showing how resources are managed.
- Write a sequence of system calls required to create and execute a new process in such an environment.

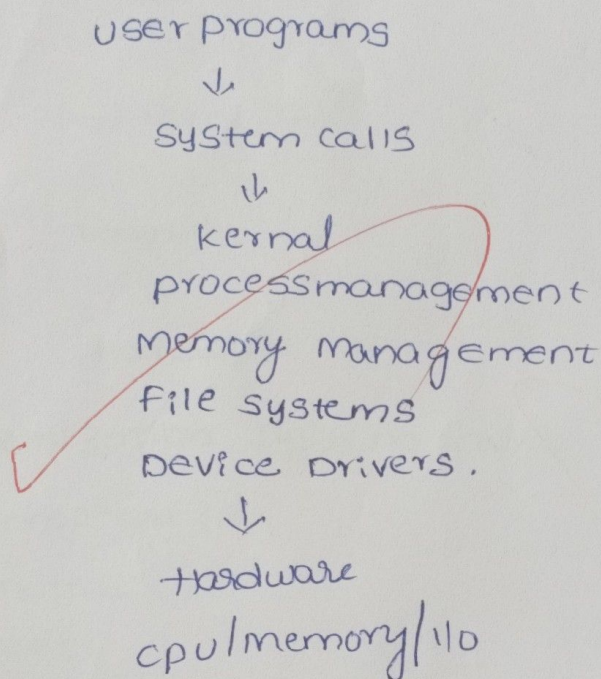
Type :

multi-user , multitasking , Timesharing operating System.

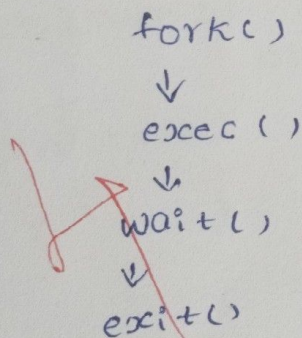
Justification :

- * supports multiple users simultaneously.
- * Executes multiple processes at the same time.
- * shares cpu, memory , and i/o devices efficiently.
- * provides fairness through cpu scheduling

Structure of an operating system



System call sequence



File and Device management System calls

File management

open()

read()

write()

close()

delete()

Device management

ioctl()

read()

write()

open()

close()

Layered structure

Application



System calls



File System



Device Drivers



Hardware

C - based System call sequence

```
int fd;
```

```
char buffer[100];
```

```
fd = open("file.txt", O_RDONLY);
```

```
read(fd, buffer, sizeof(buffer));
```

```
close(fd);
```

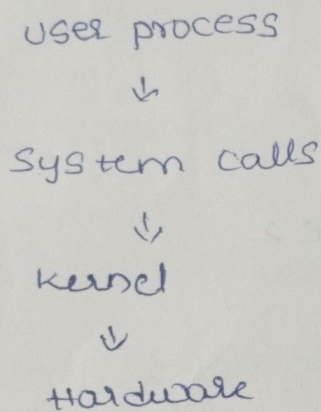

Suitable operating system

Real time operating systems (RTOS)

Why Suitable

- * fast response
- * predictable execution
- * Meets deadlines
- * Reliable for industrial control.

Interactive Diagram



Synchronization System calls

- * Semaphore (,)
- * mutex (,)
- * wait+()
- * signal (,)

purpose:

prevent race conditions and synchronize processes.

Suitable operating System

Distributed operating System.

Comparison

Distributed OS
Multiple computers
Resources sharing
High reliability
Better scalability

centralized OS
single computer
limited sharing
Less reliable
less scalable.

Distributed OS Architecture.

Node 1 $\longleftrightarrow$ Network $\longleftrightarrow$ Node 2

↑
Kernel
↑
Applications

↑
Kernel
↑
Applications

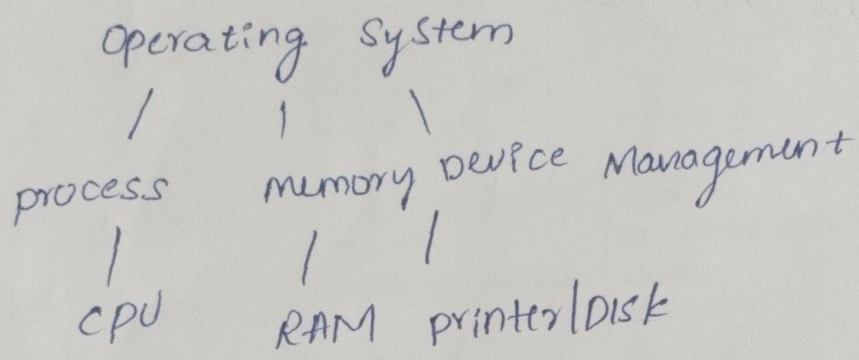
IPC System call sequence

Pipe()
fork()
write()
read()
close()

major functions of OS

- * process scheduling
- * Memory management
- * Device management
- * file management
- * security and protection

b) Block Diagram



c) System call Examples

Memory Management

- * `brk()`
- * `mmap()`
- * `munmap()`

Device Management

- * `open()`
- * `read()`
- * `write()`

* close()

* ioctl()

Conclusion:

An operating system efficiently manages processes, memory, files, and devices while providing secure and reliable communication between users, applications and hardware.

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RUBRICS FOR CALCULATION

Criteria	Weightage (Marks)	Level 4 – Exemplary (Full Marks)	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	35 Marks	Demonstrates clear, complete, and in-depth understanding of concepts (types, structure, system calls).	Good understanding with minor gaps. Basic understanding with some misconceptions.	Basic understanding with some misconceptions.	Poor or incorrect understanding
Accuracy of Answers	25 Marks	All answers are correct, precise, and relevant.	Mostly correct with minor errors	Partially correct with noticeable error	Mostly incorrect or irrelevant.
Application of Knowledge	15 Marks	Effectively applies concepts to scenarios/questions.	Applies concepts with minor mistakes	Limited or partial application	No meaningful application
Completeness of Response	15 Marks	All parts of each question are fully answered.	Most parts covered with small omissions	Some parts missing	Incomplete answer.
Clarity and Presentation	10 Marks	Clear, well-structured, and neatly presented answers	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand and poorly presented